Suivi du comportement de la faune à l'aide de la localisation acoustique

Lucas Voirin, André Desrochers, Marc J. Mazerolle





UNIVERSITÉ LAVAL
Faculté de foresterie, de géographie et de géomatique

lucasvoirin.github.io/CEF_2026



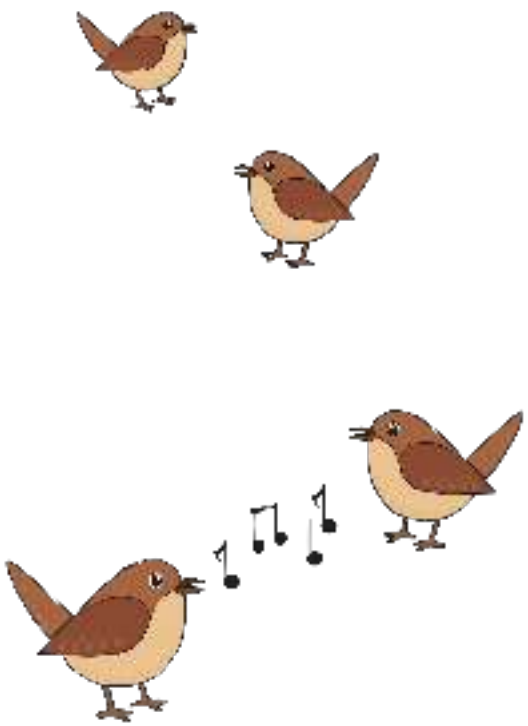
Évaluer des impacts



Les enjeux



Les enjeux



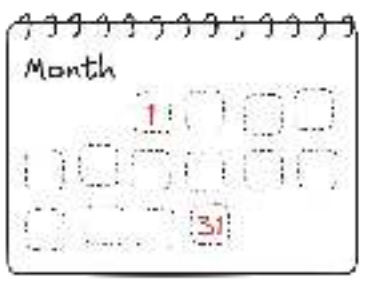
Les enjeux



Les enjeux



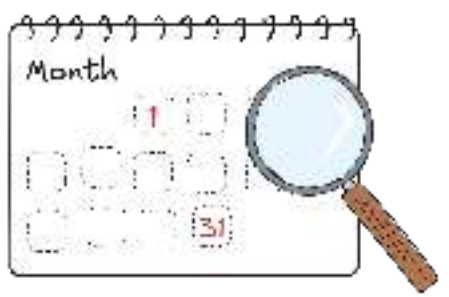




Les enjeux





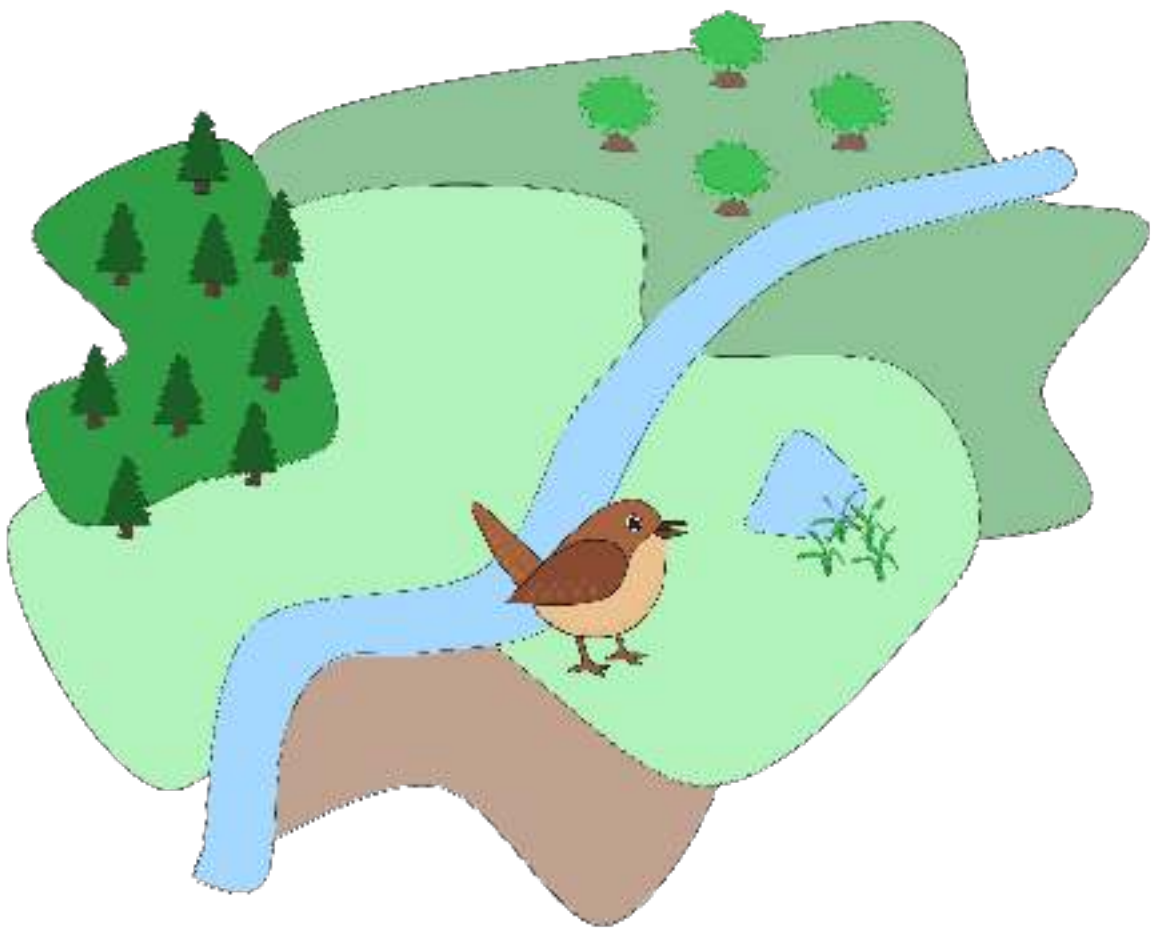


Le comportement

Mennill et al. (2006), Dutilleux et al. (2023), Teixeira et al. (2019)

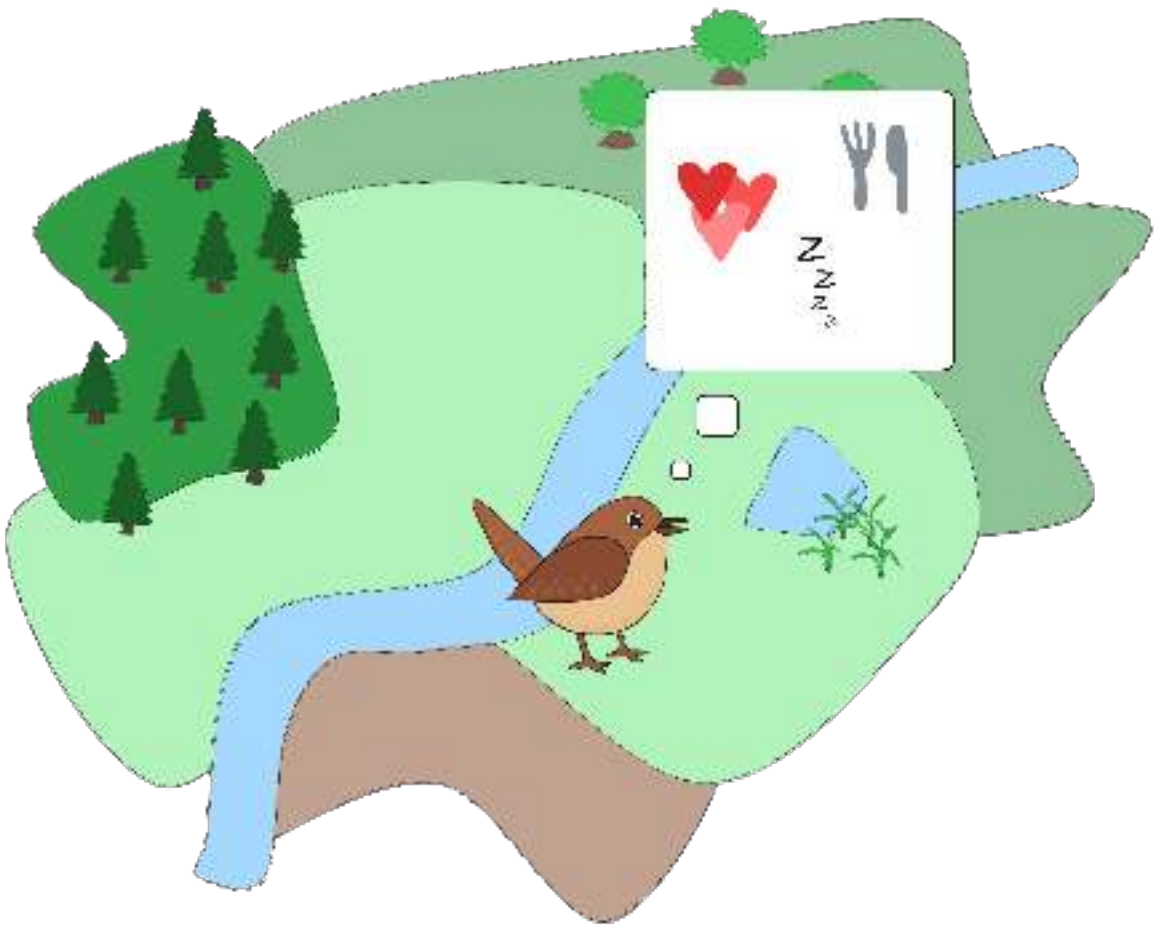


Le comportement



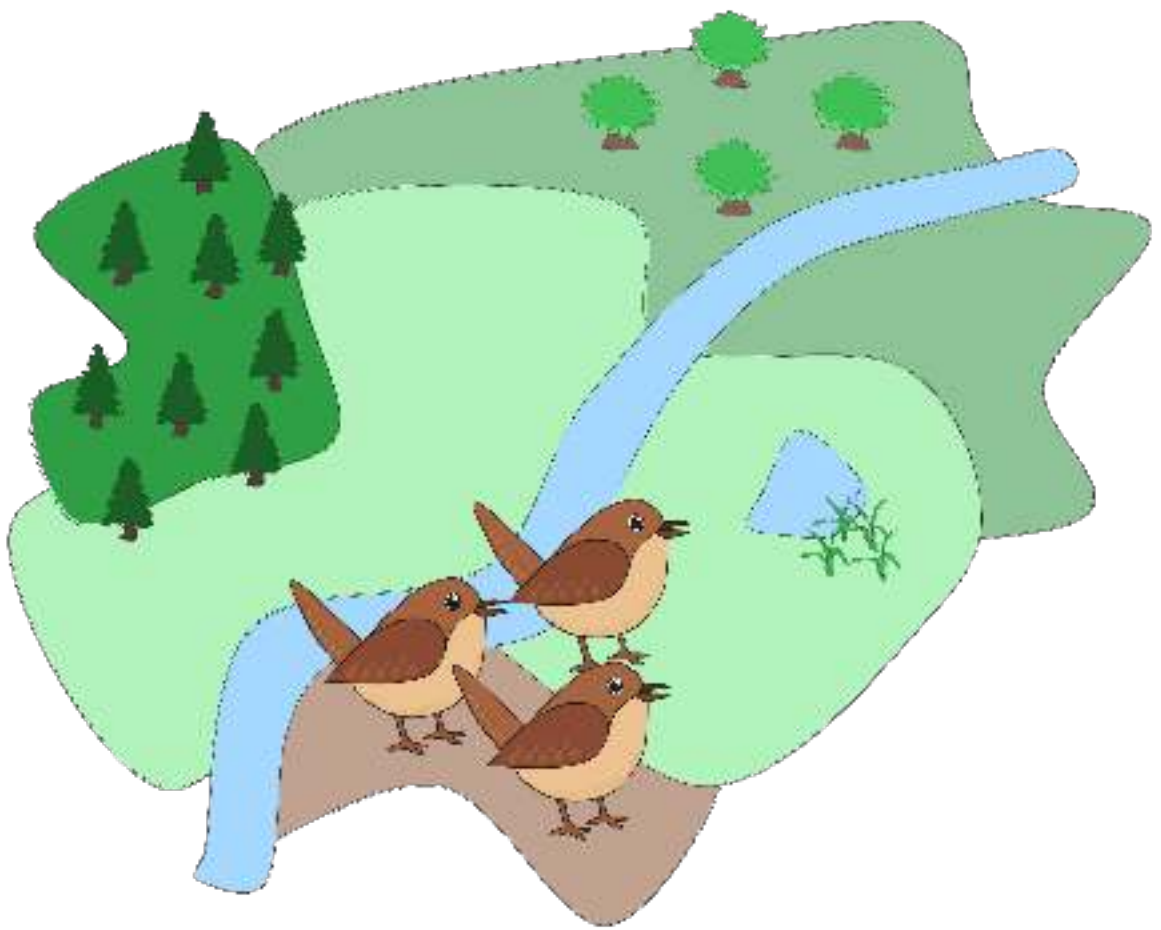
Mennill et al. (2006), Dutilleux et al. (2023), Teixeira et al. (2019)

Le comportement



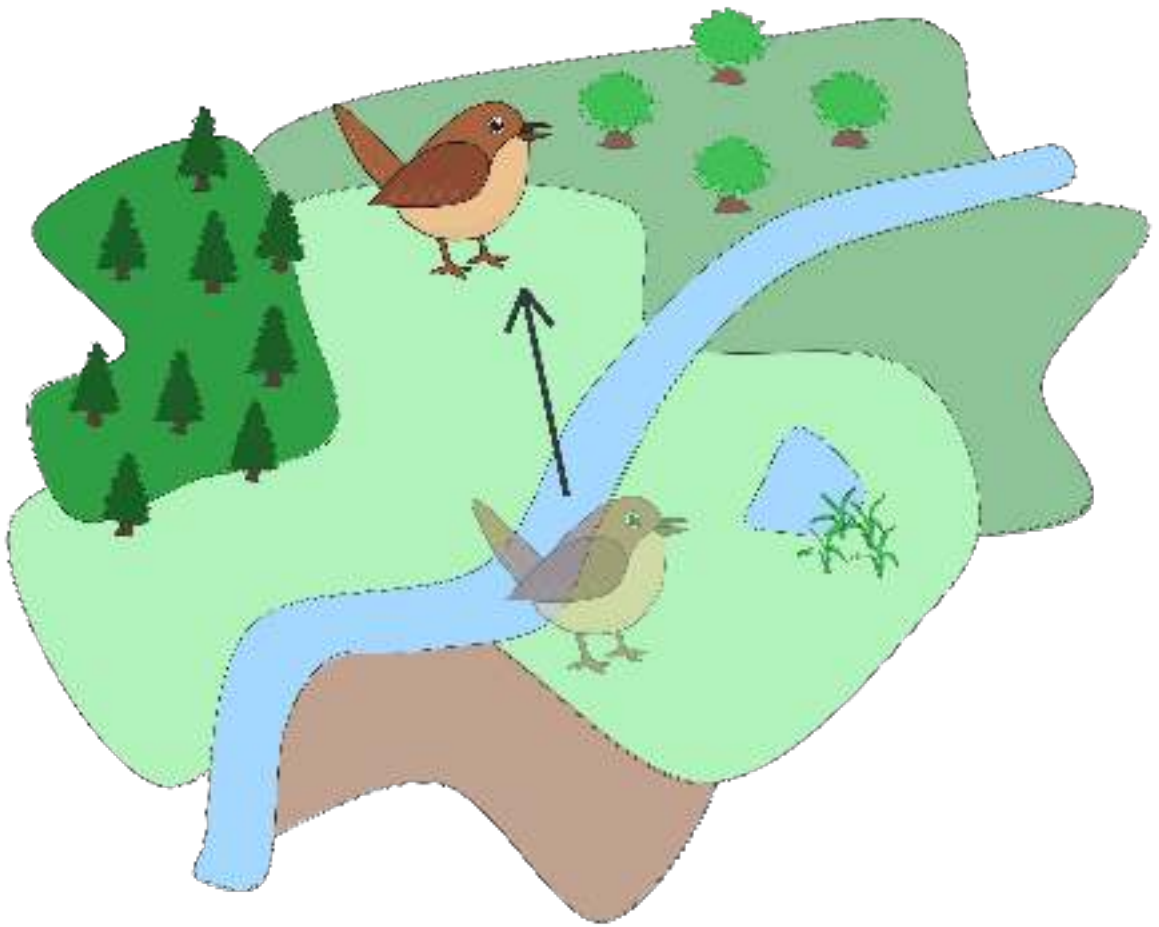
Mennill et al. (2006), Dutilleux et al. (2023), Teixeira et al. (2019)

Le comportement



Mennill et al. (2006), Dutilleux et al. (2023), Teixeira et al. (2019)

Le comportement



Mennill et al. (2006), Dutilleux et al. (2023), Teixeira et al. (2019)

Le suivi acoustique passif

Gibb et al. (2019), Sugai et al. (2019)



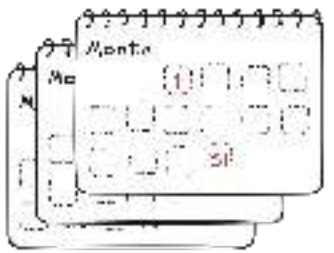
Le suivi acoustique passif



Gibb et al. (2019), Sugai et al. (2019)

Le suivi acoustique passif

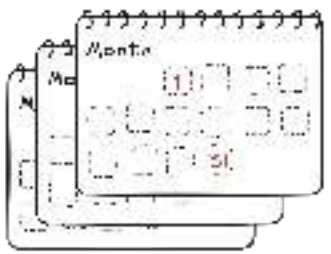




Gibb et al. (2019), Sugai et al. (2019)

Le suivi acoustique passif



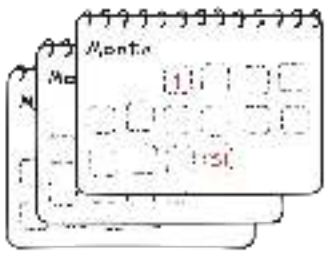




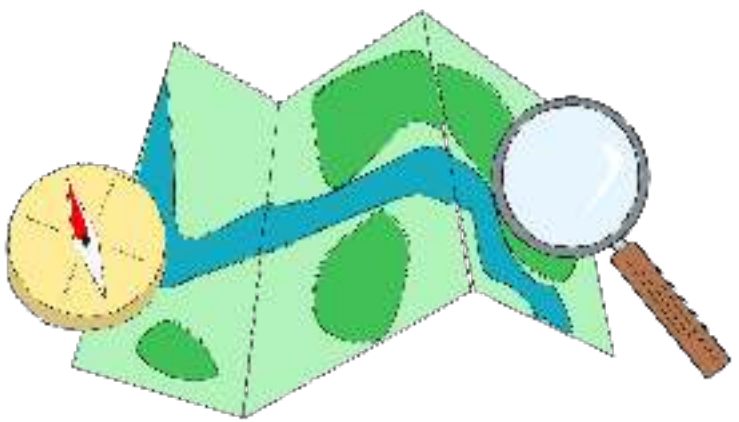
Gibb et al. (2019), Sugai et al. (2019)

Le suivi acoustique passif









Gibb et al. (2019), Sugai et al. (2019)

Localiser des sons



Sur le terrain

Rhinehart et al. (2020)



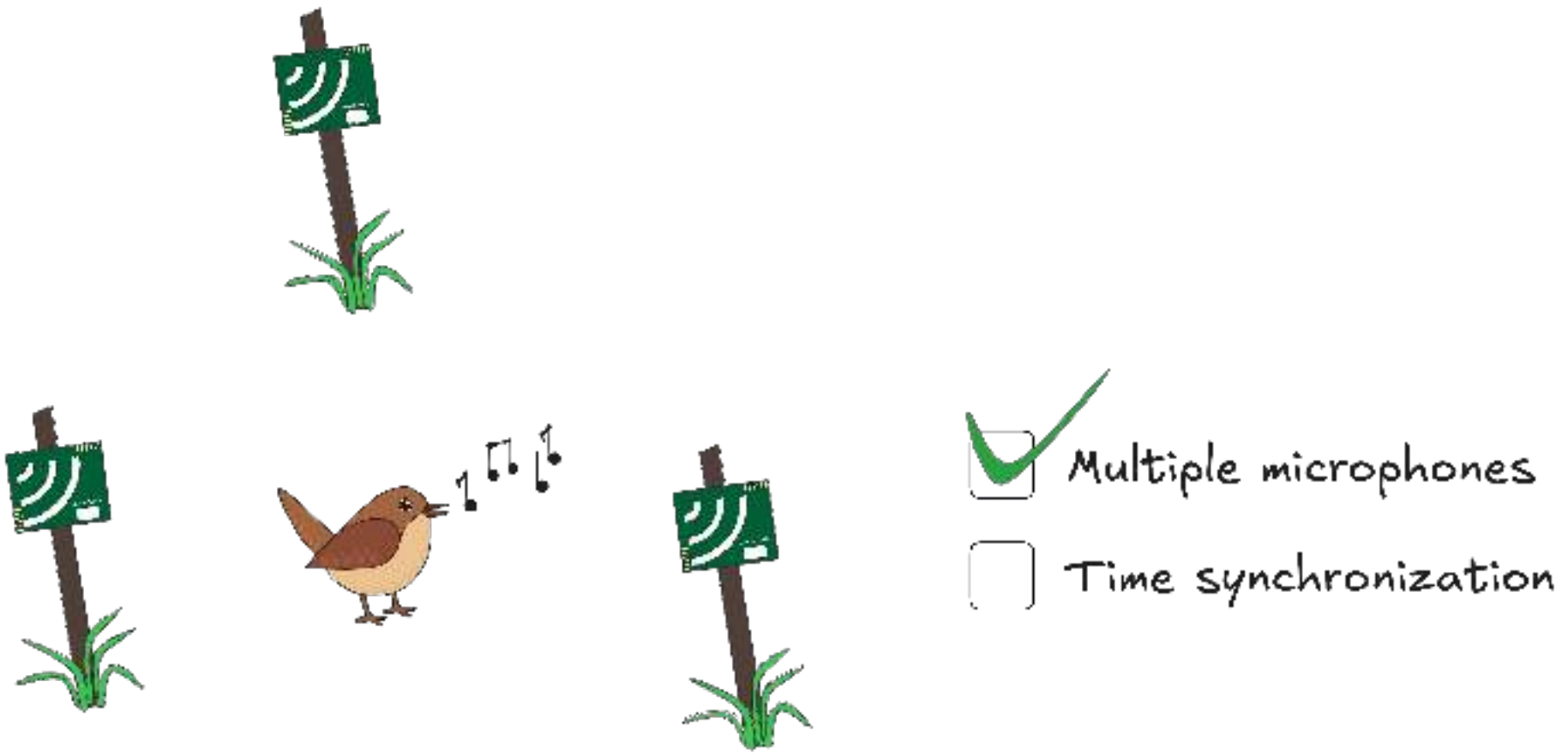
Sur le terrain



- [ ] Multiple microphones
- [ ] Time synchronization

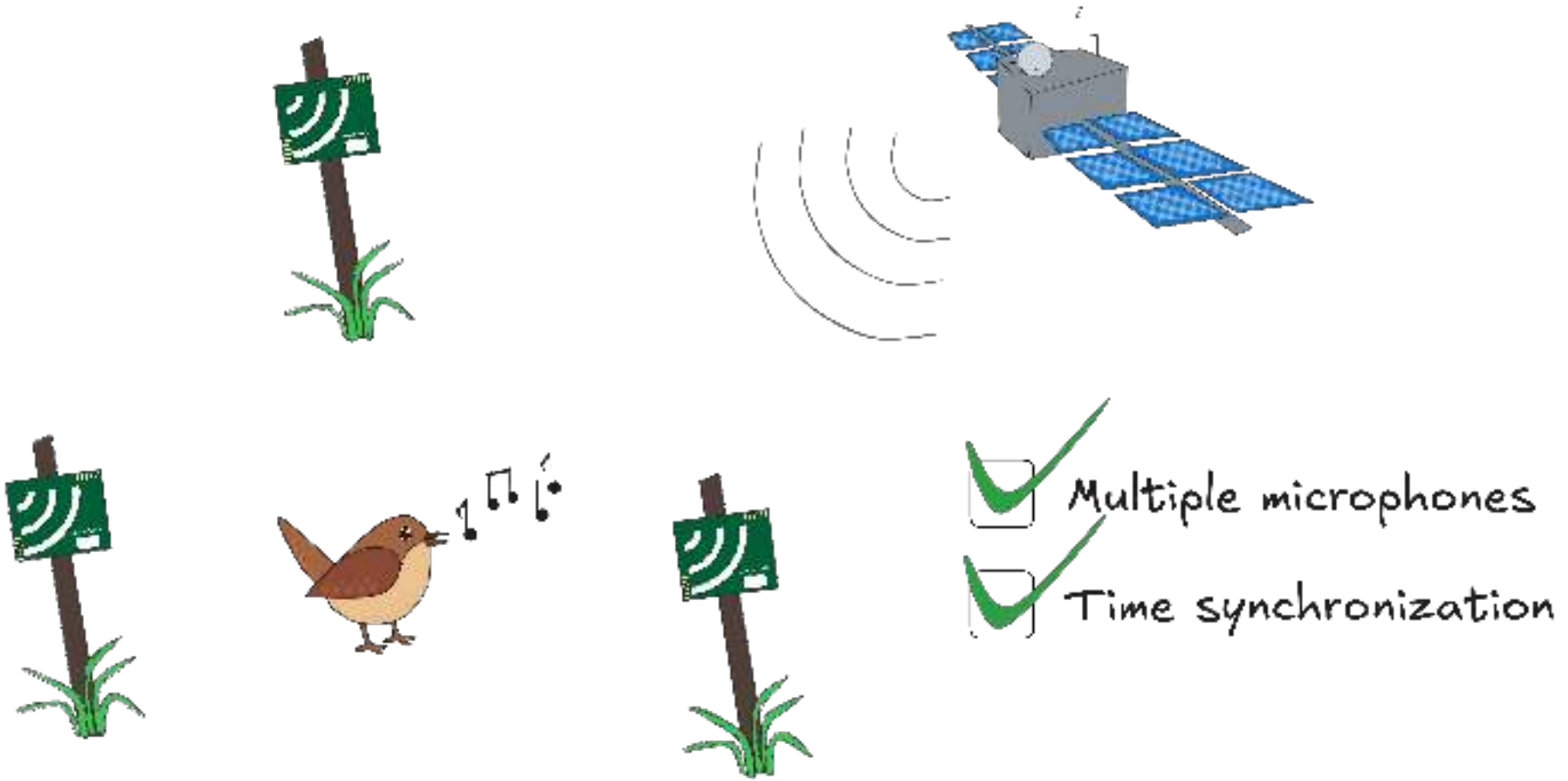
Rhinehart et al. (2020)

Sur le terrain



Rhinehart et al. (2020)

Sur le terrain



Rhinehart et al. (2020)

Objectifs



Objectifs

Localiser précisément des sons d’animaux



Objectifs

Localiser précisément des sons d’animaux

Prendre en compte les incertitudes dans:

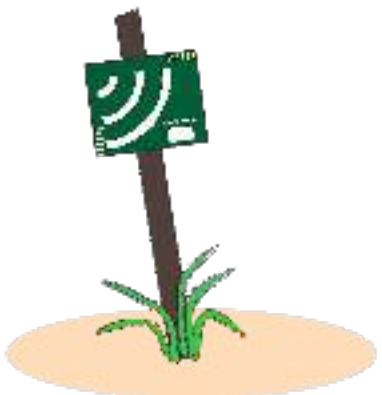
- La position des enregistreurs
- Les estimations de délais de temps d’arrivée



Au labo



Au labo

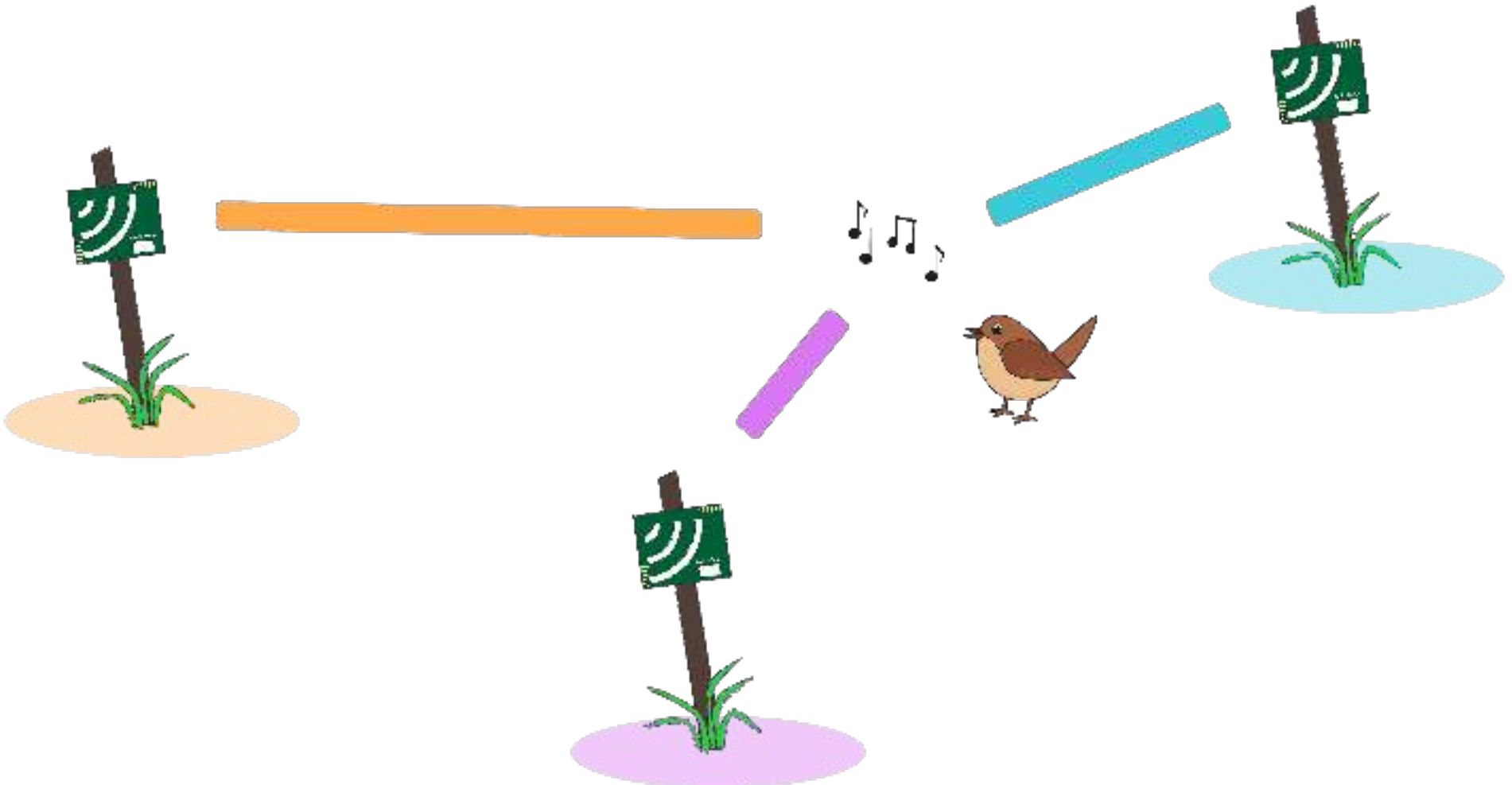




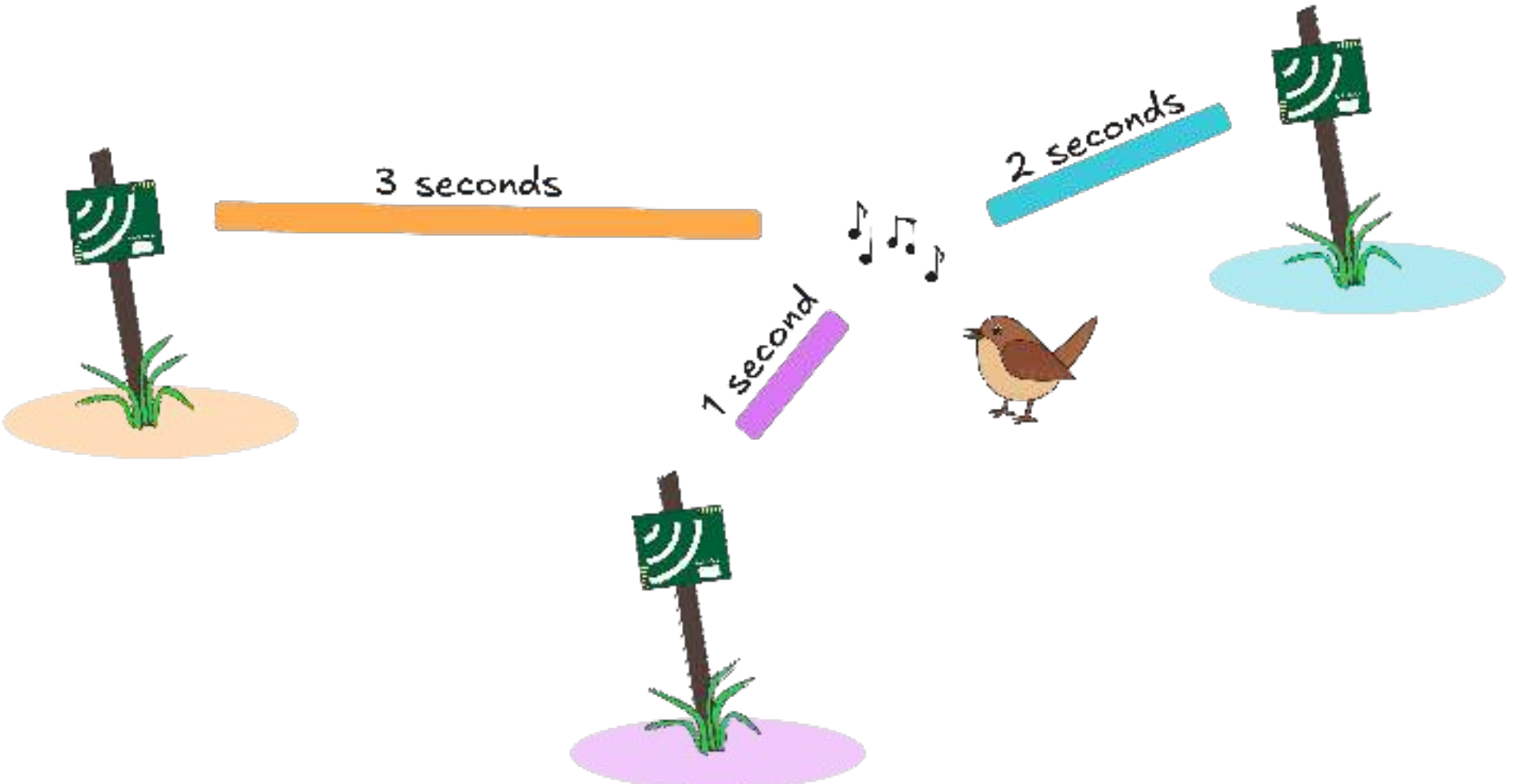




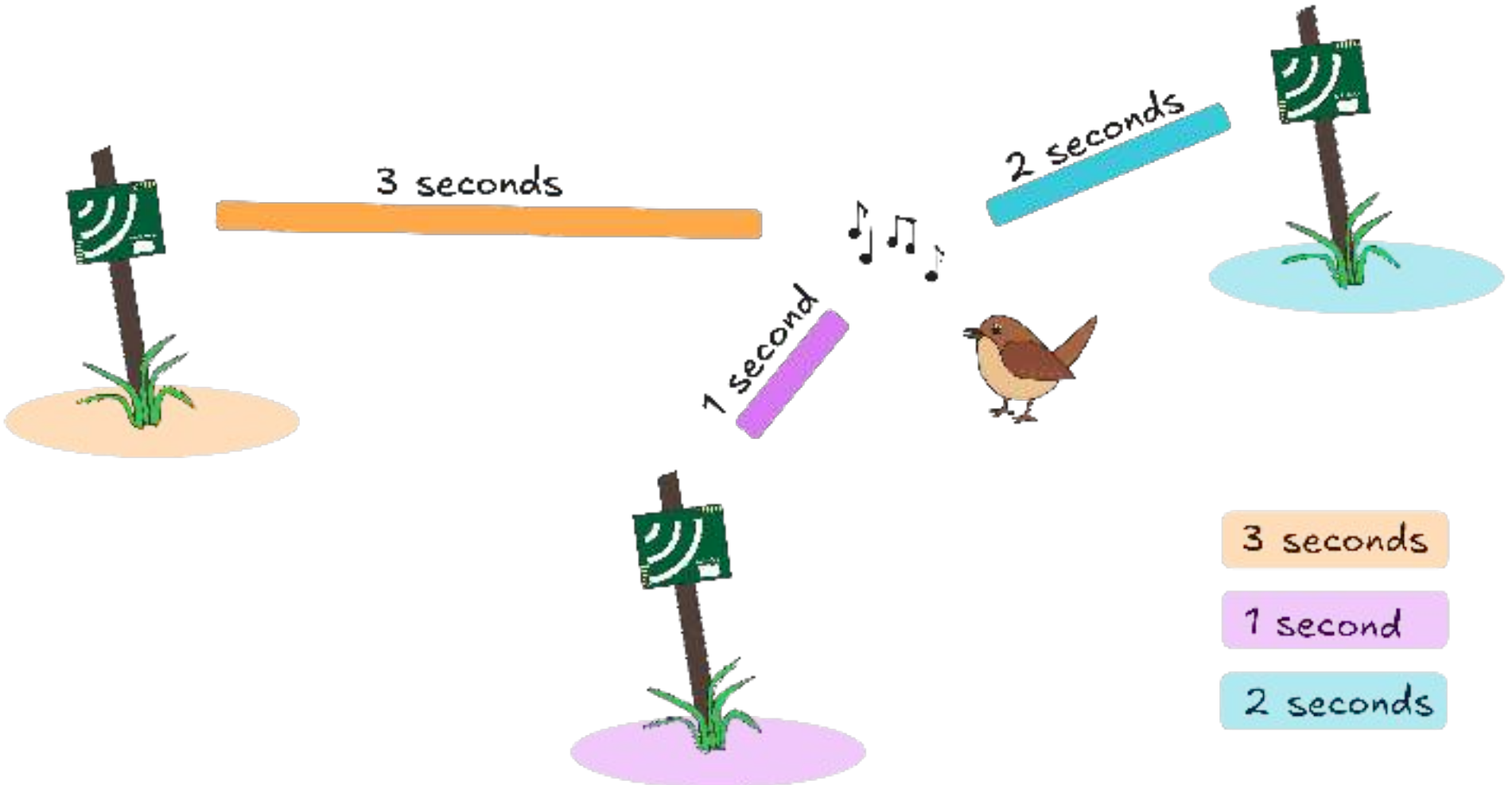
Au labo



Au labo

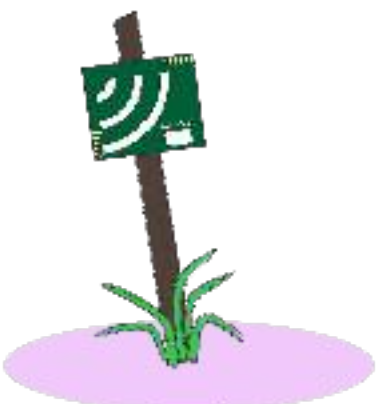


Au labo



Au labo





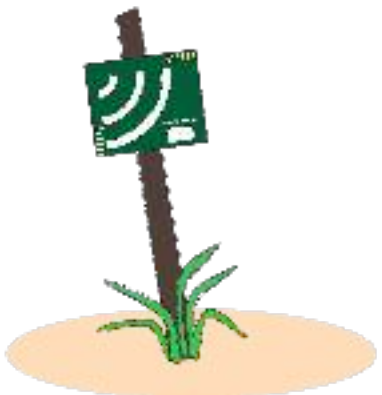


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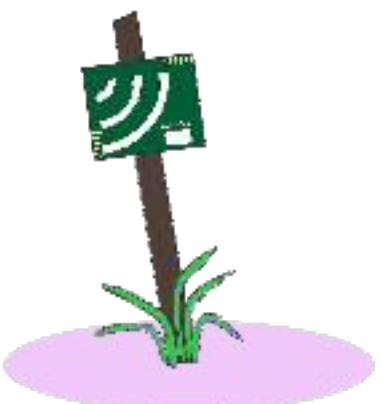
1 second

2 seconds

Au labo



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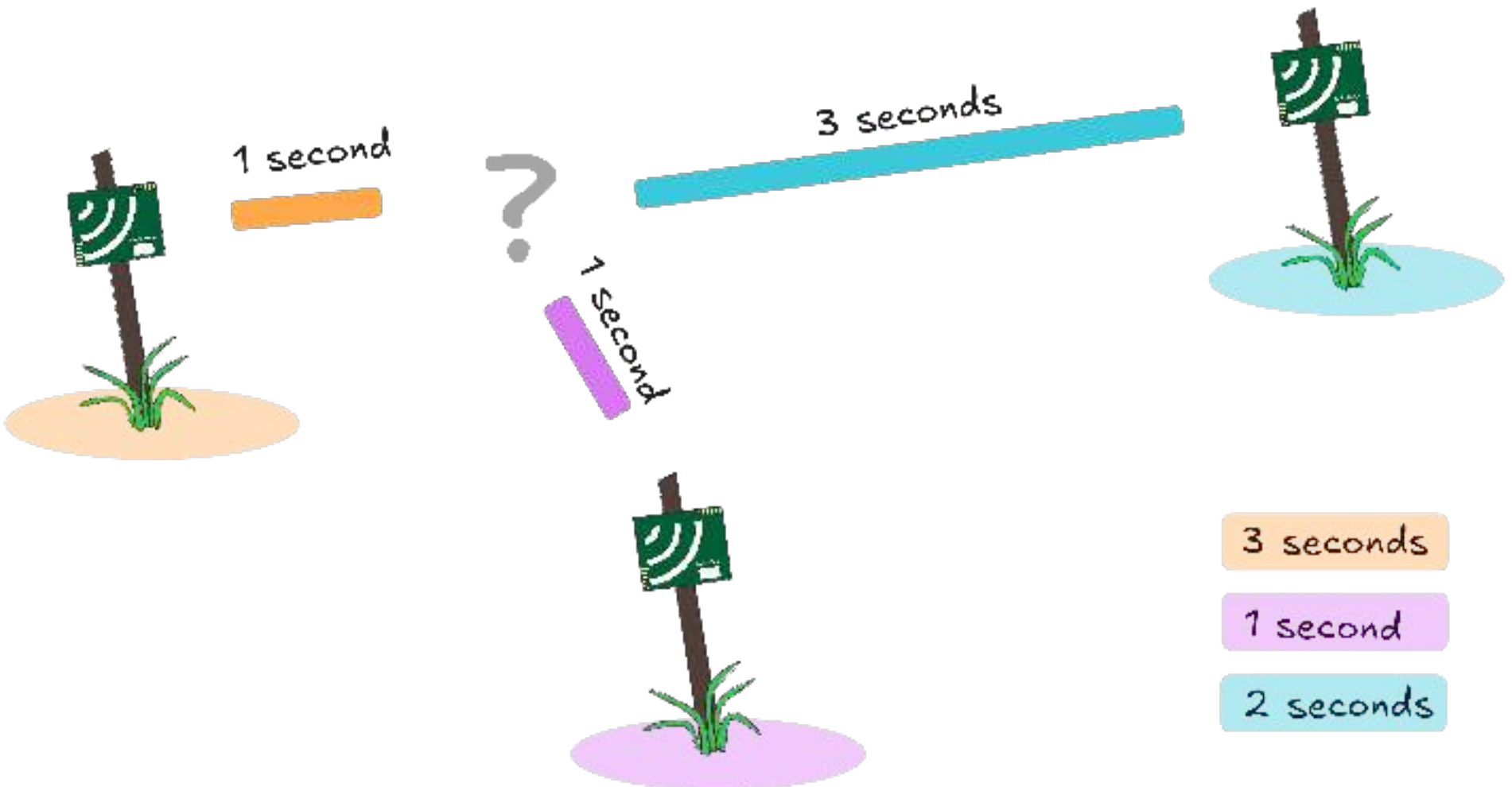


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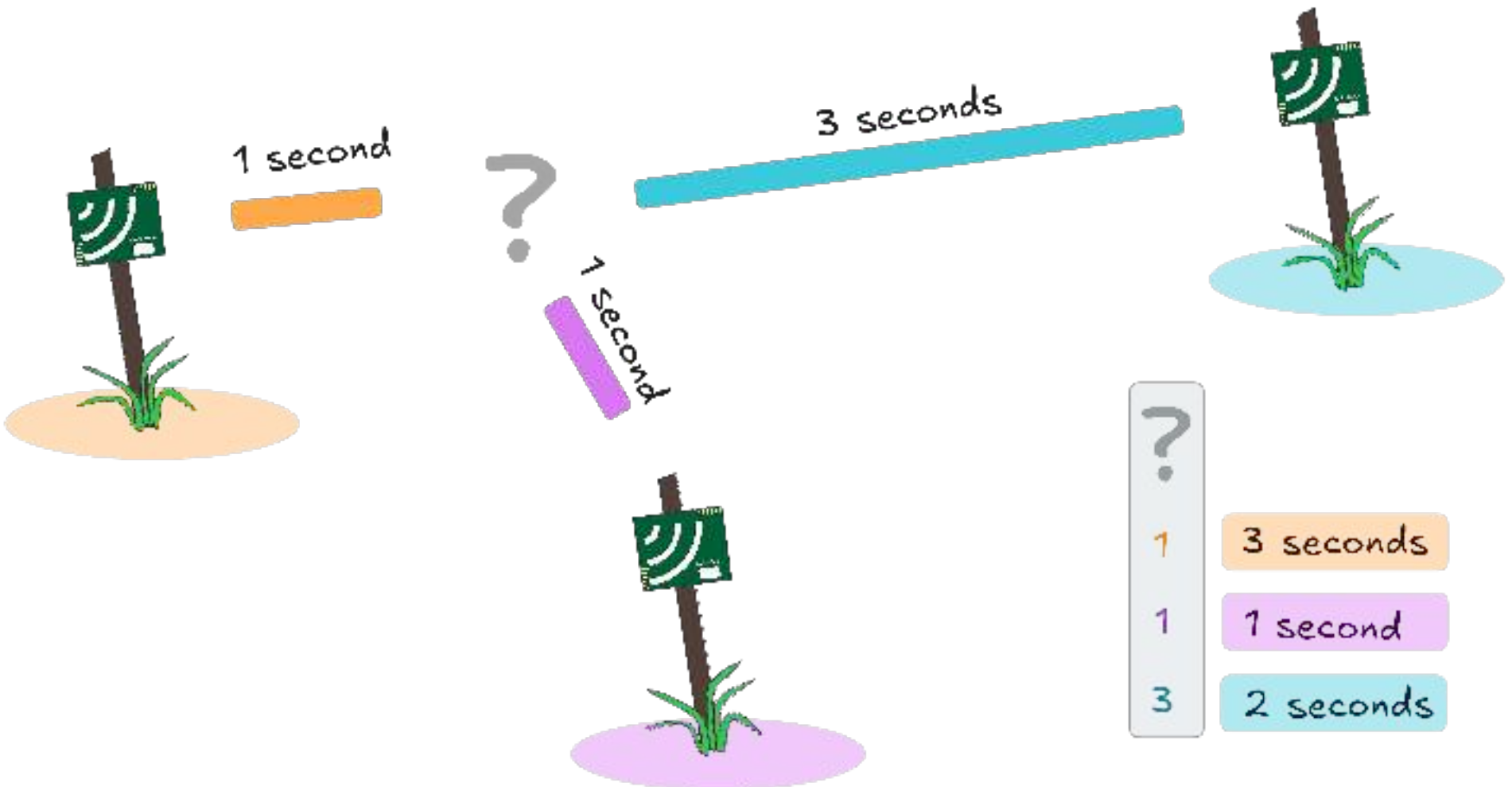
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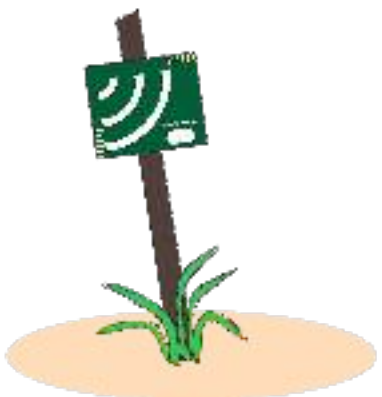
Au labo

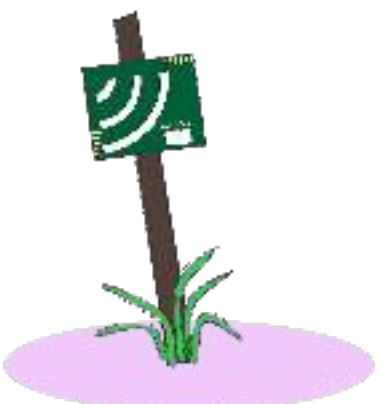


Au labo



Au labo





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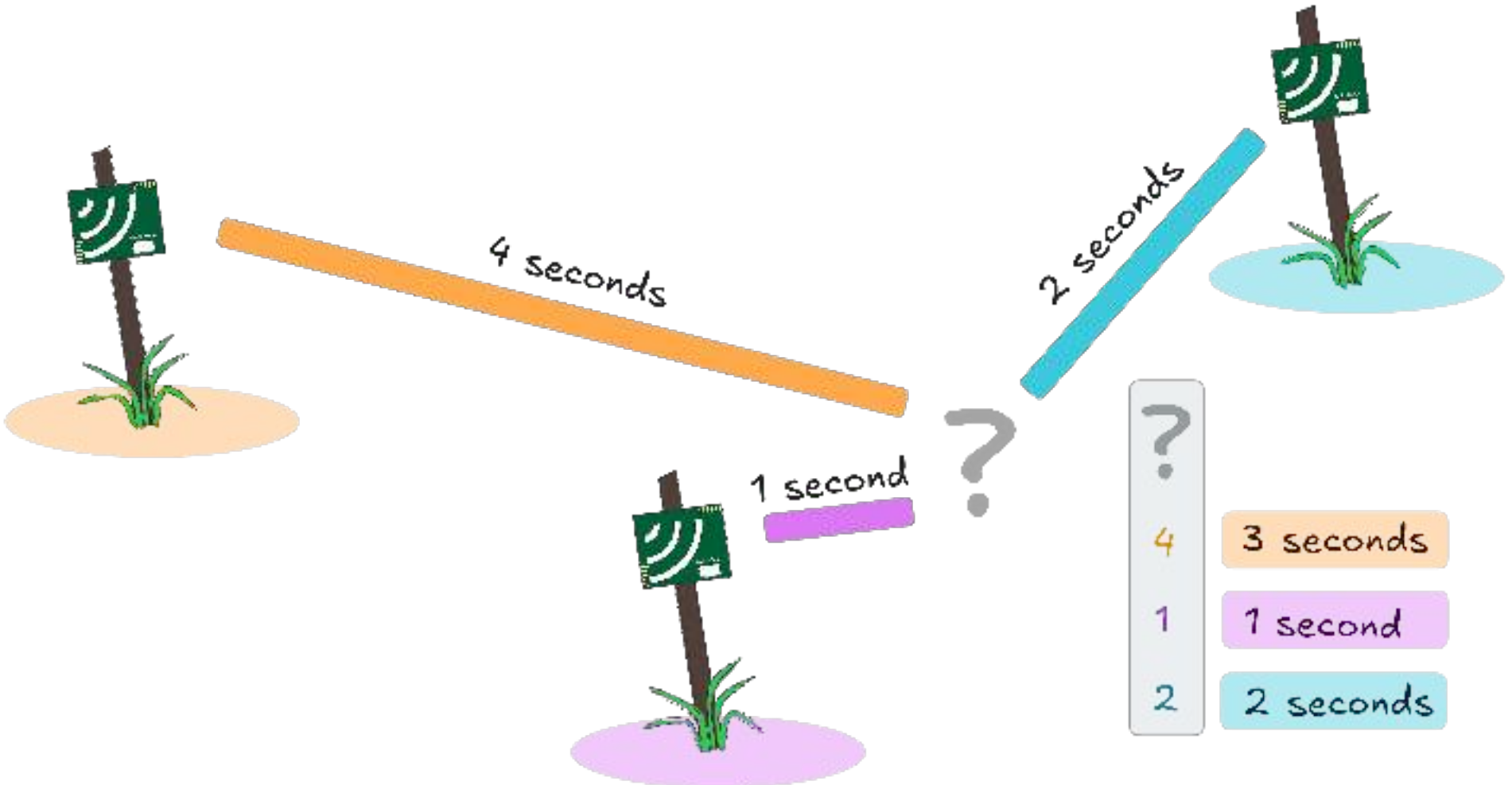
3 seconds

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2 seconds



Au labo



Au labo



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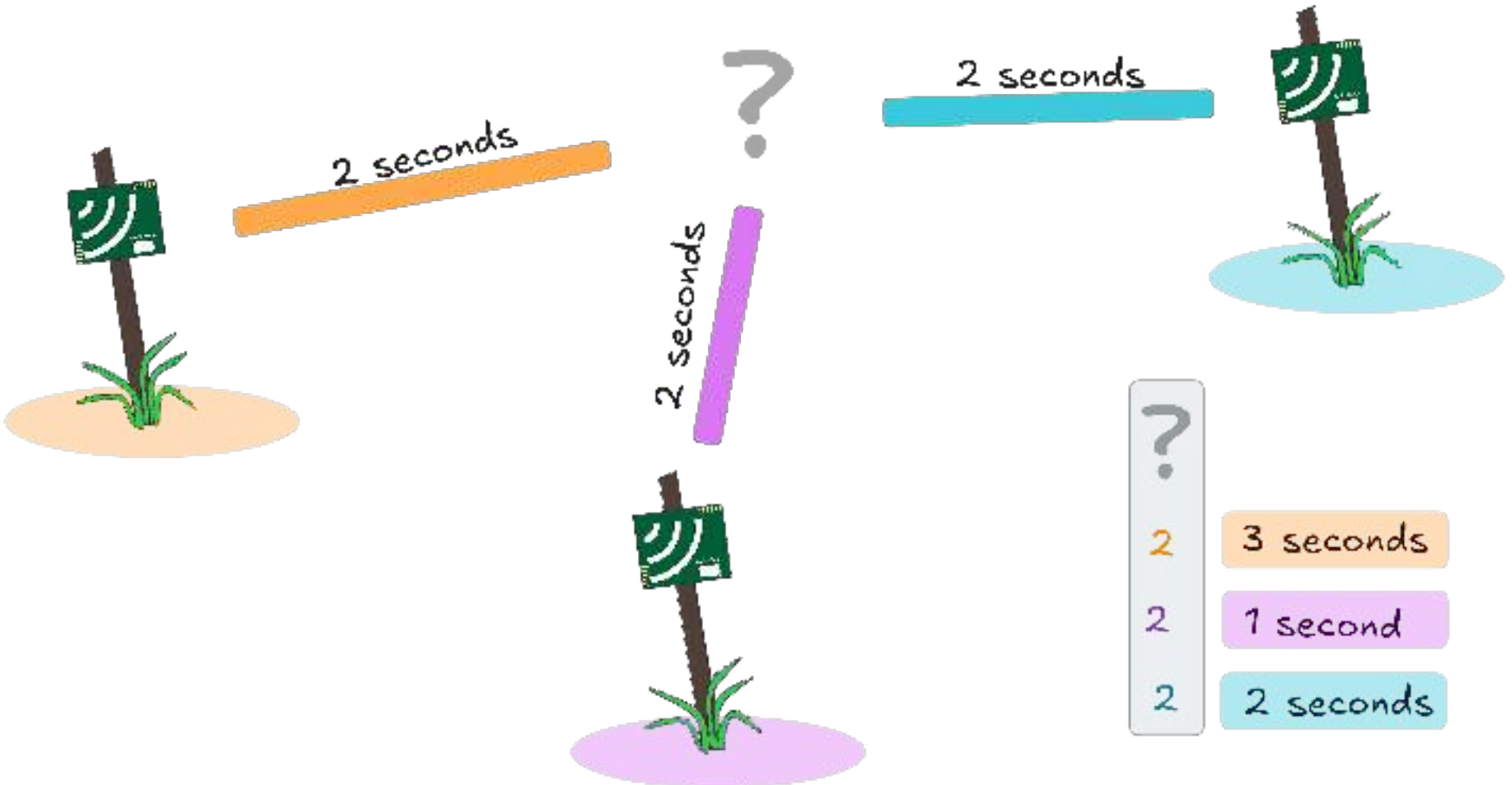
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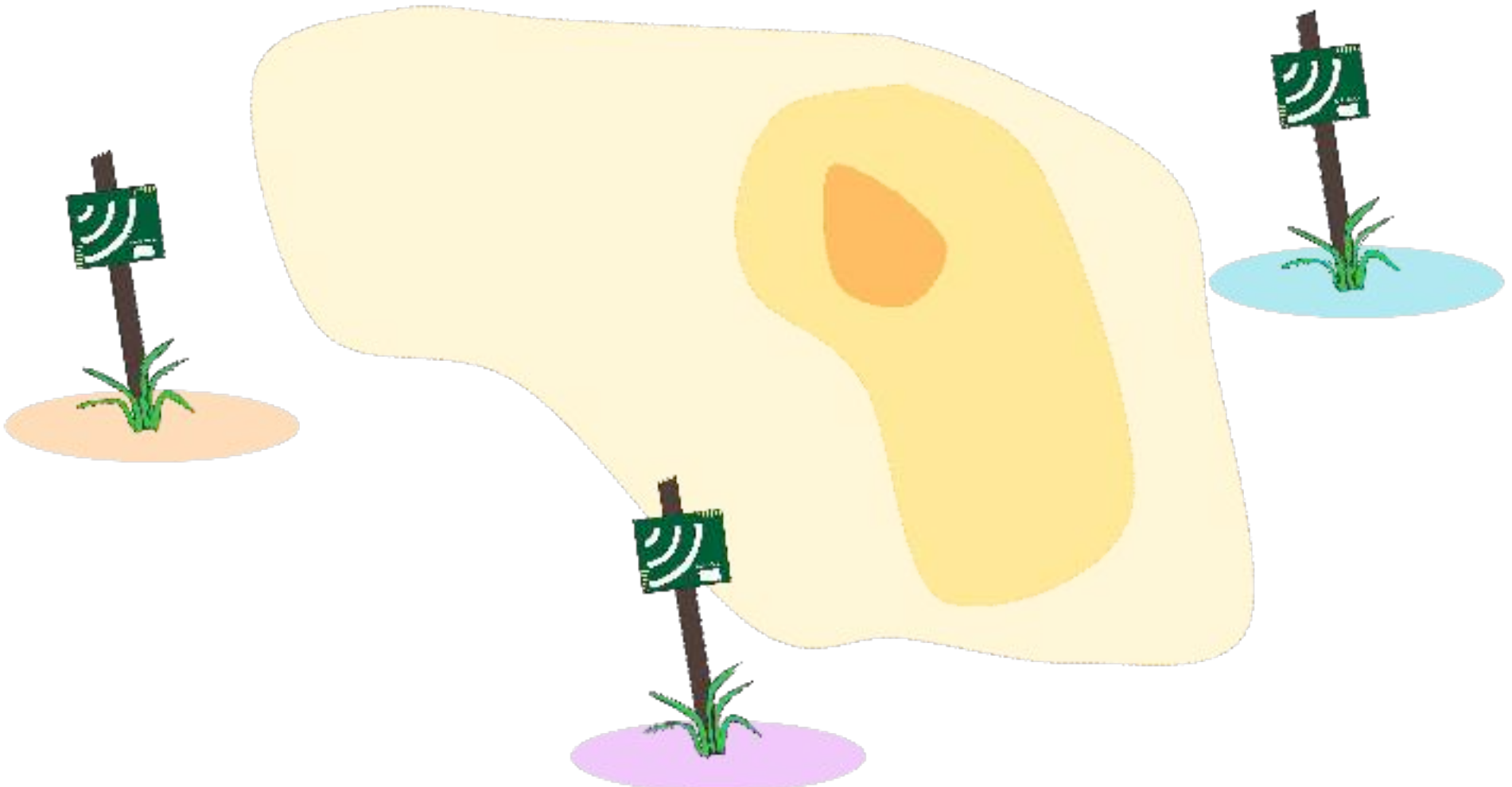
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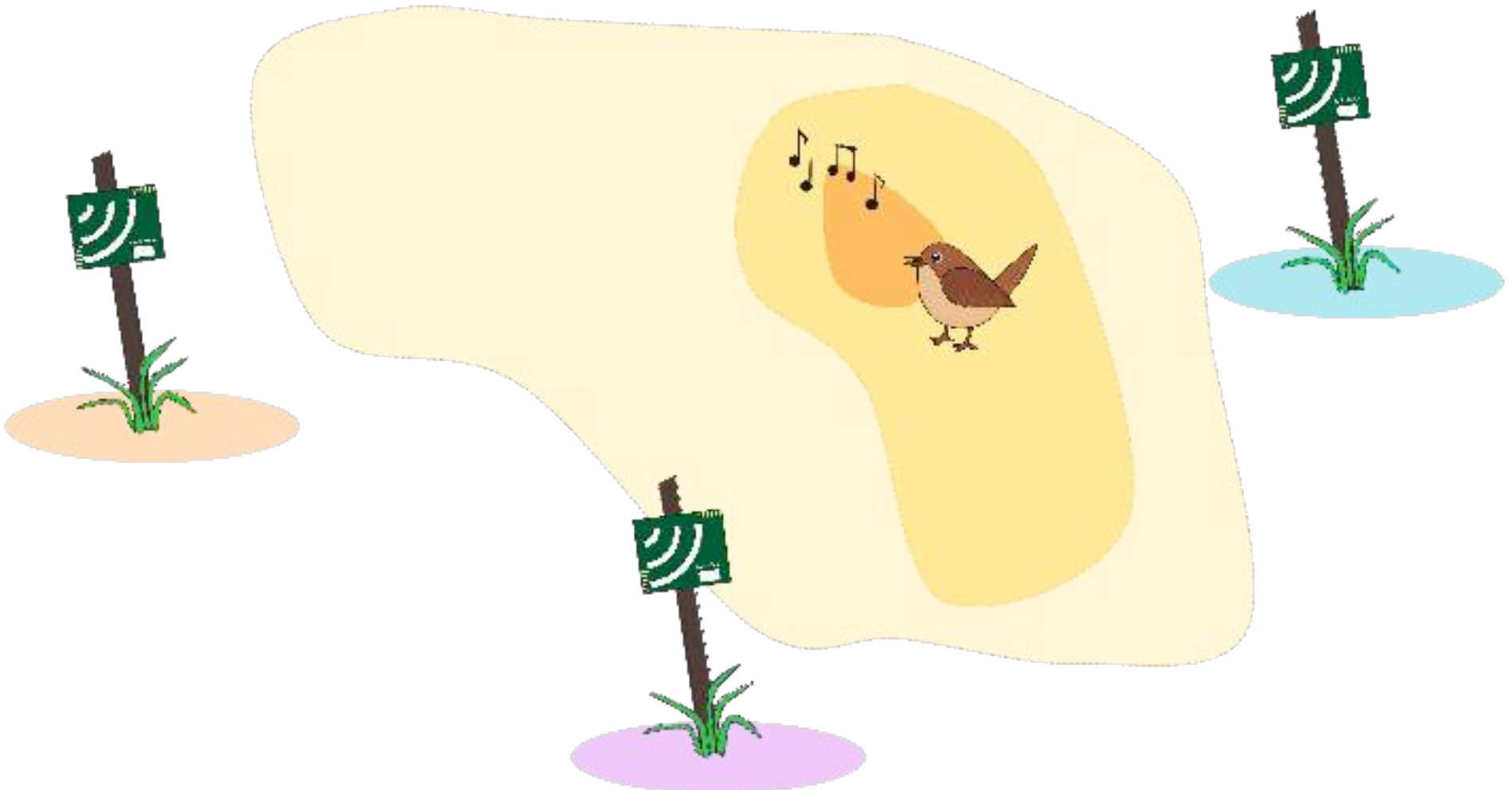
Au labo


2 seconds
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Au labo



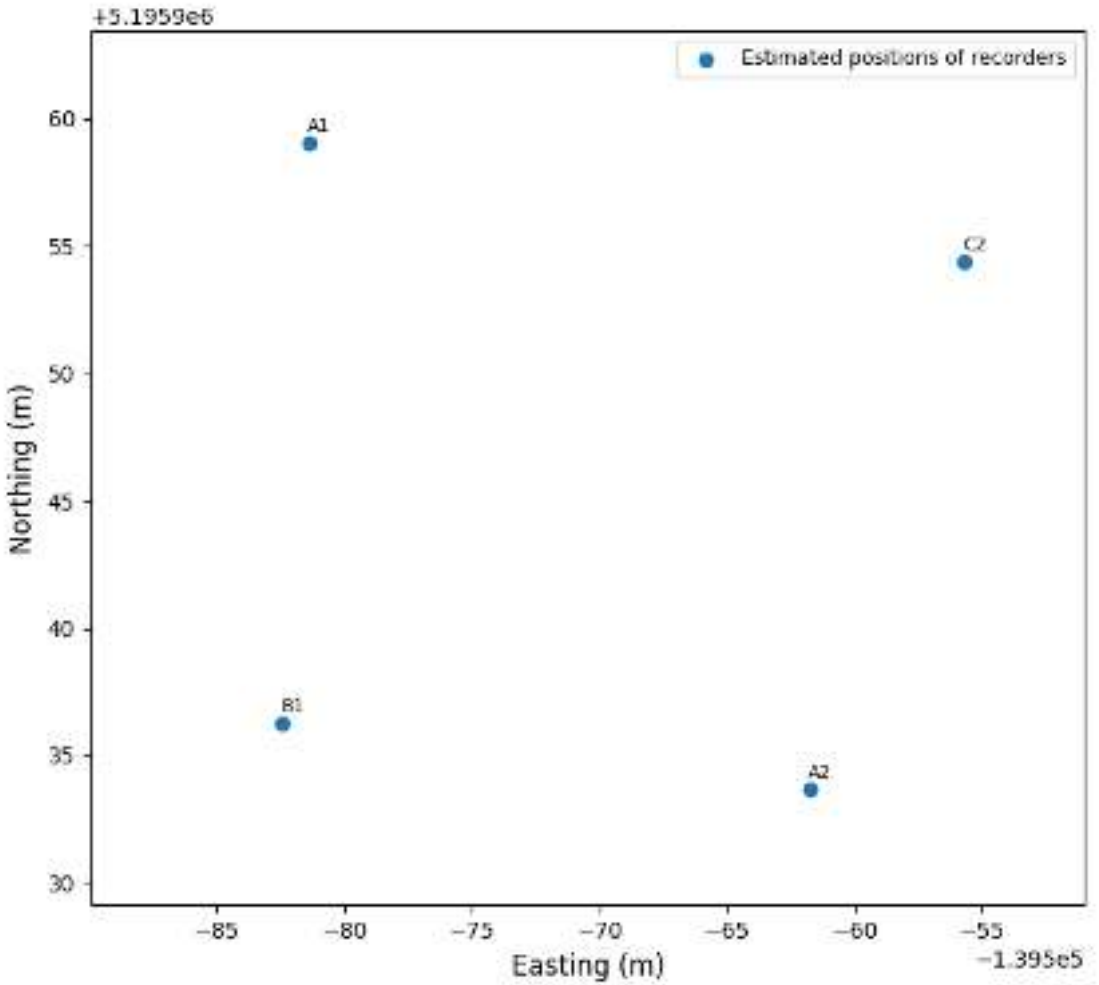
Au labo



Localisation de source sonore

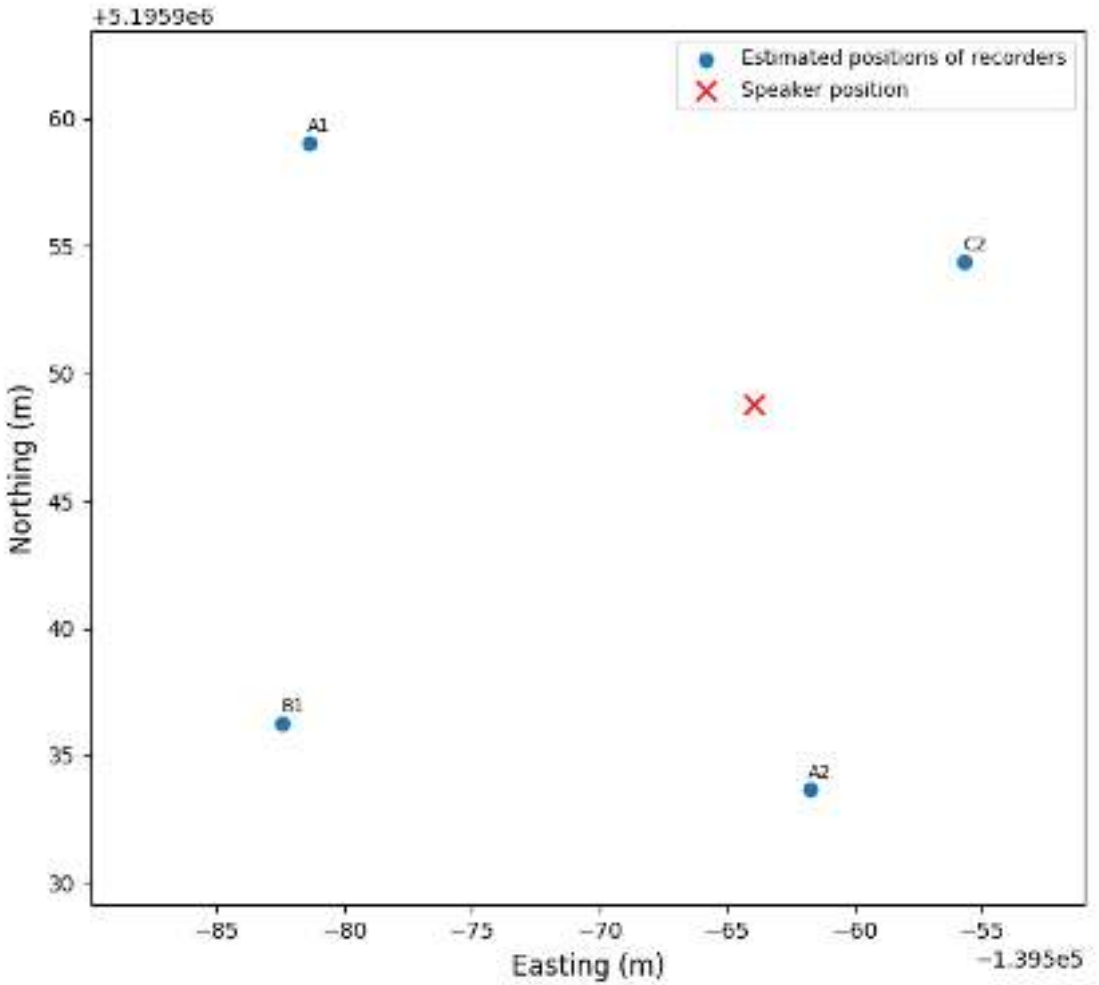


Localisation de source sonore



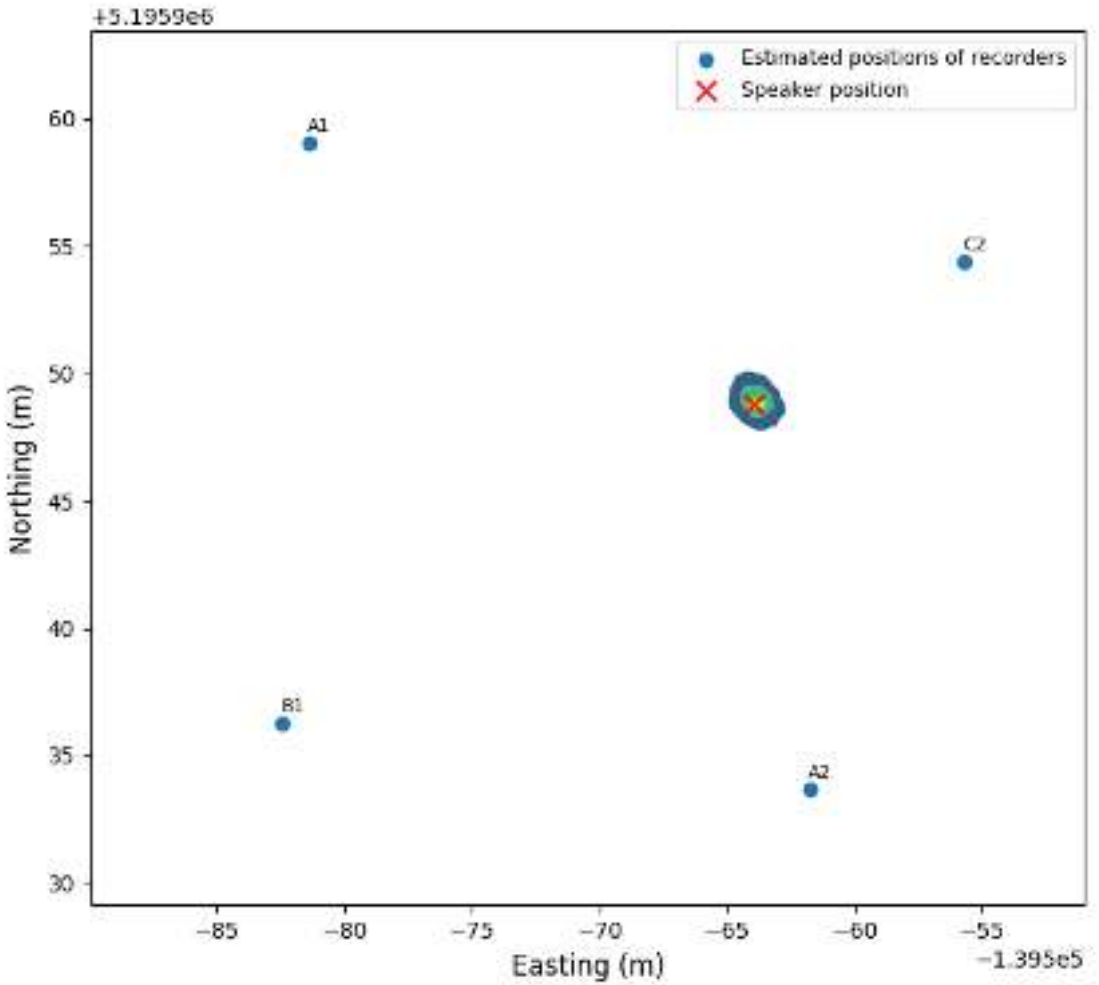
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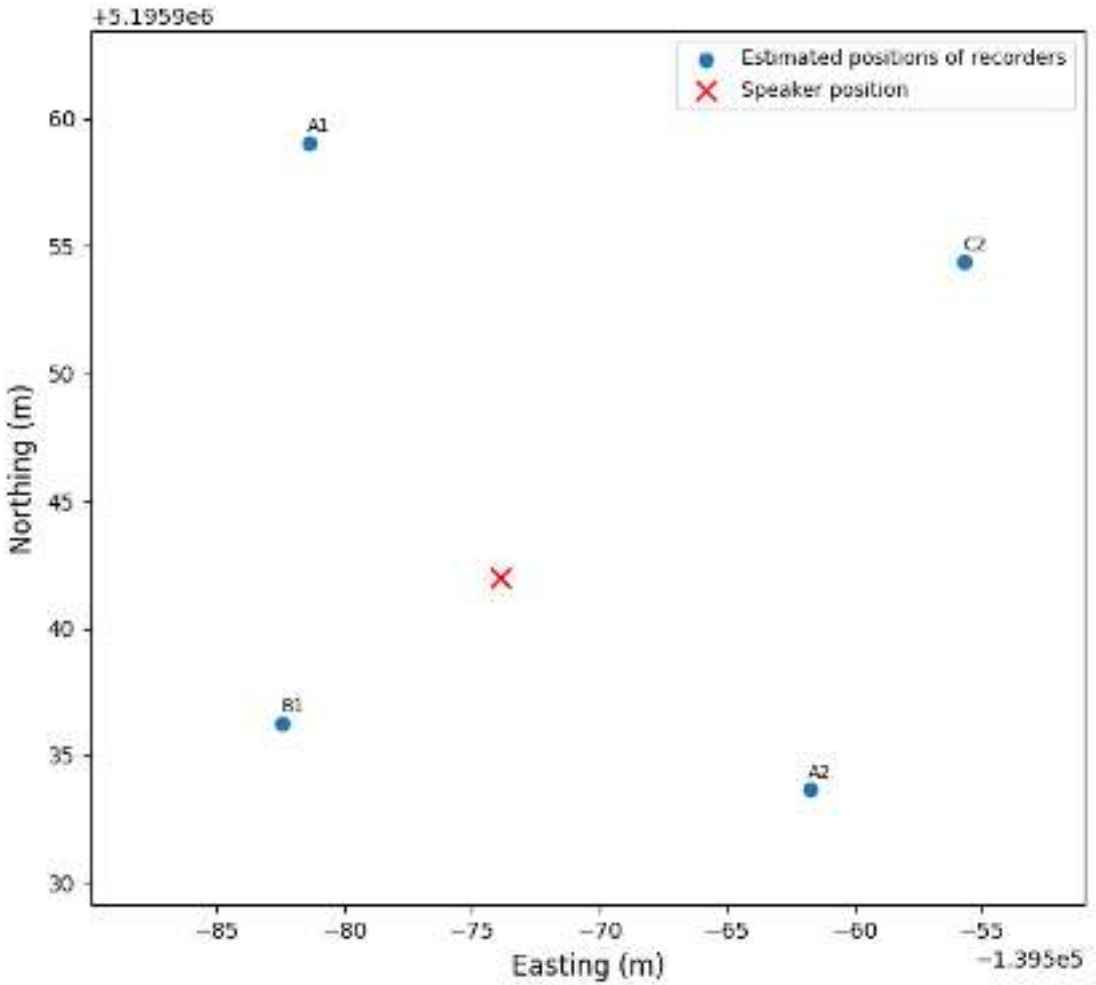
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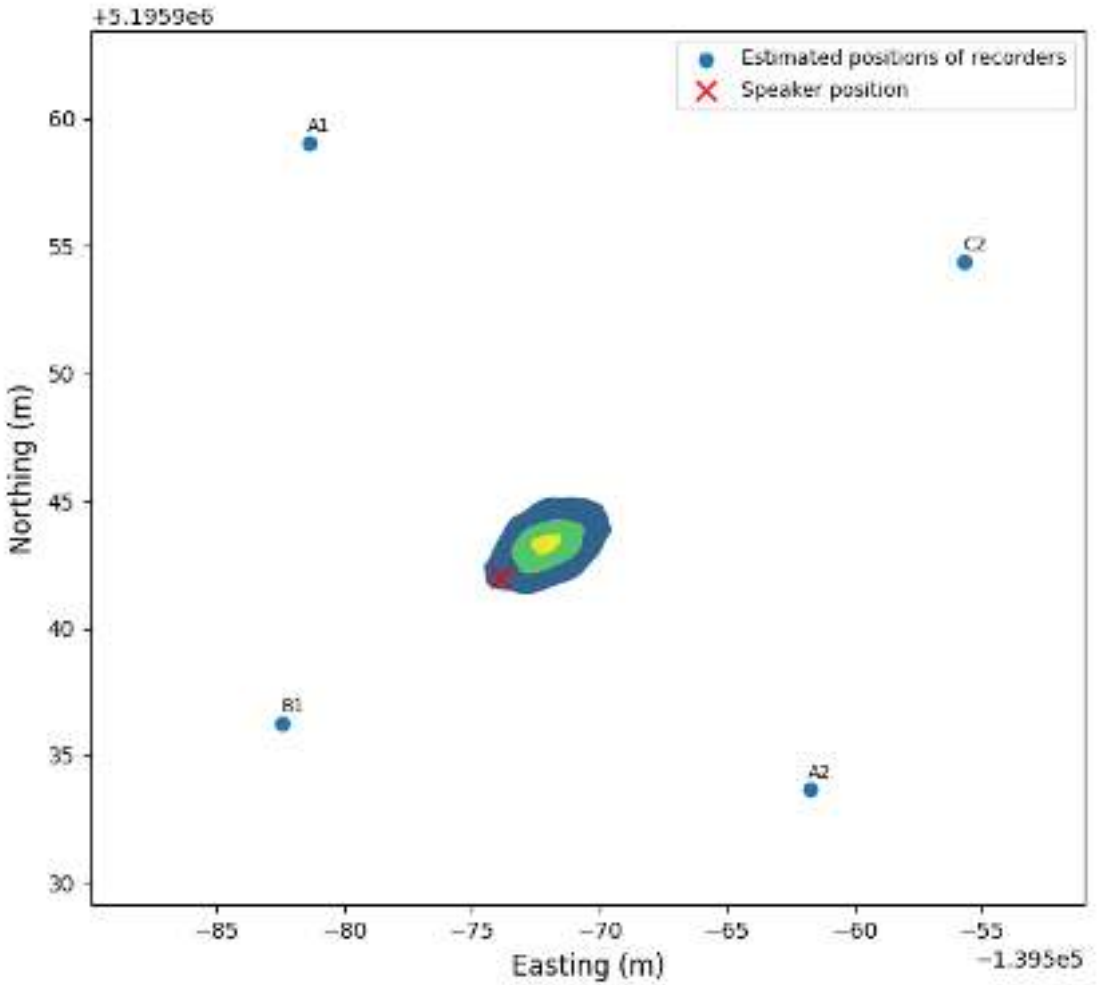
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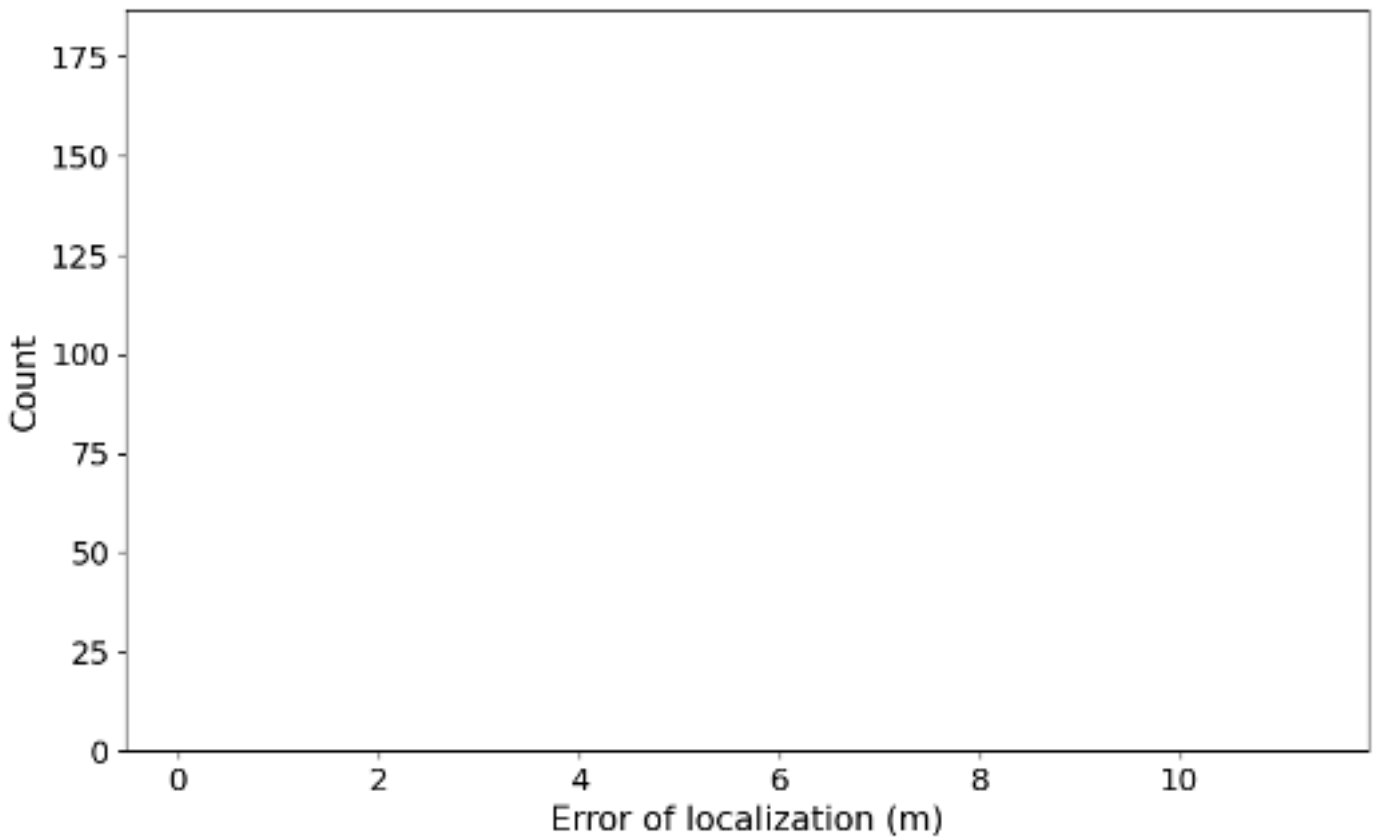


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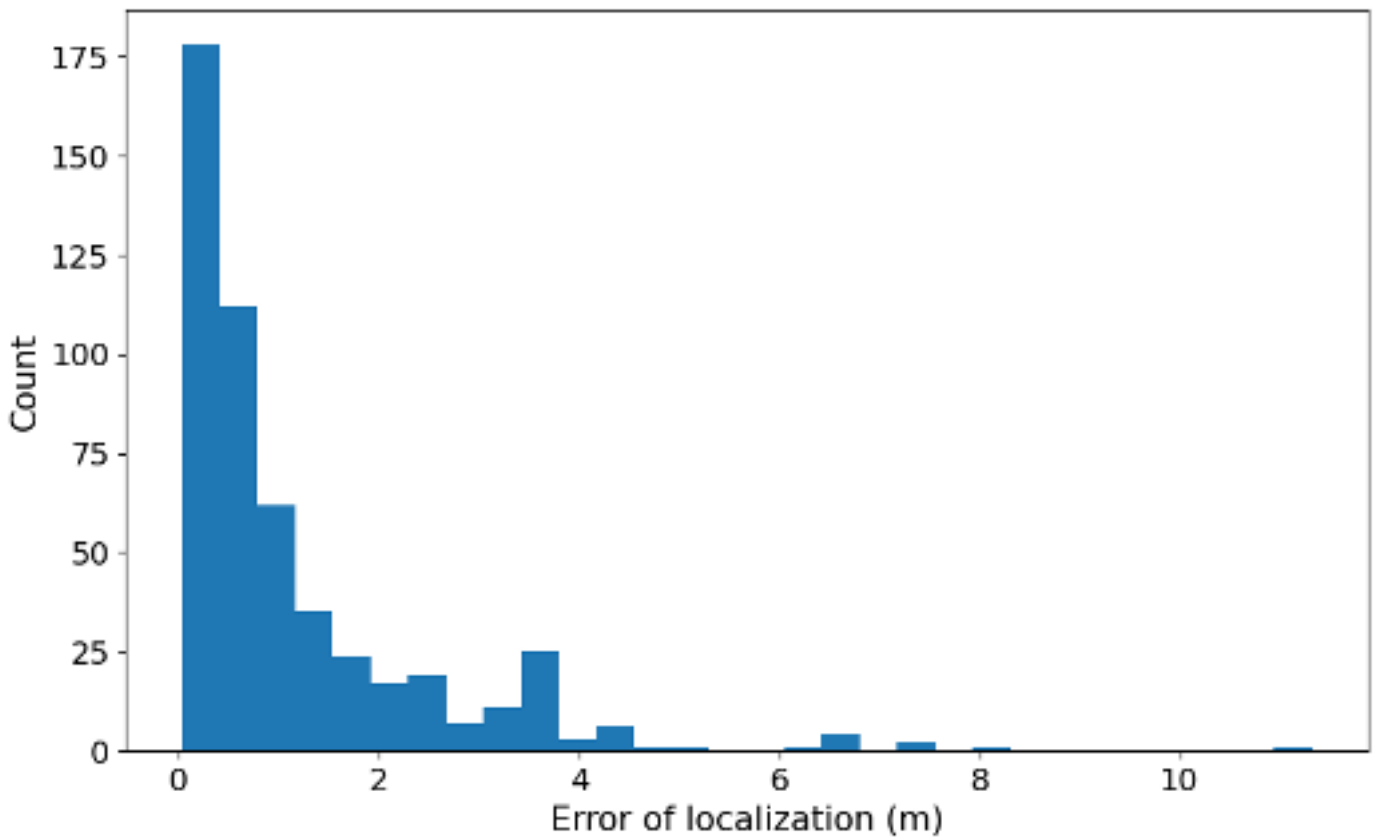
Évaluer la précision



Évaluer la précision



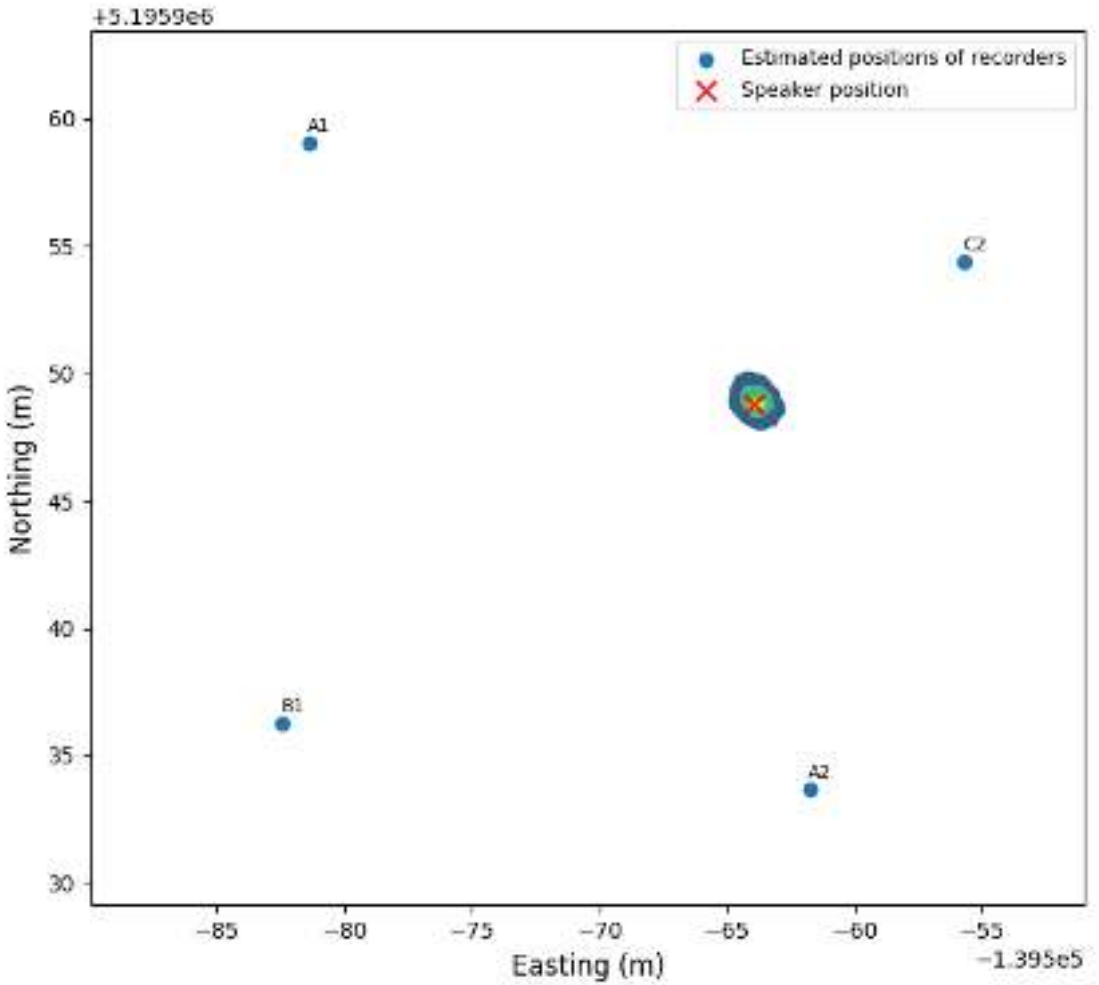
Évaluer la précision



Localiser les enregistreurs

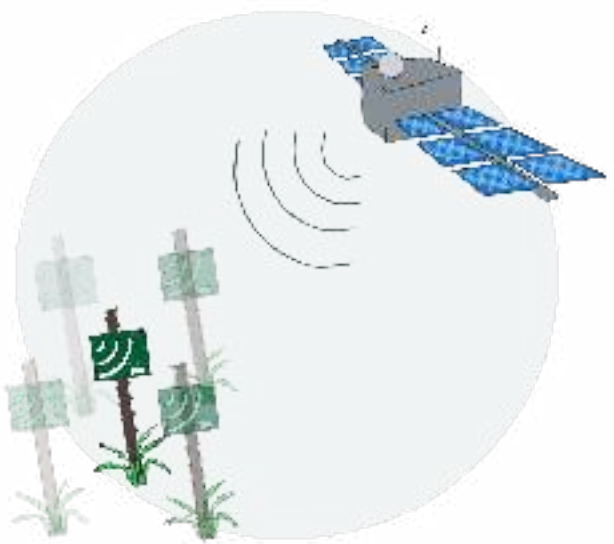


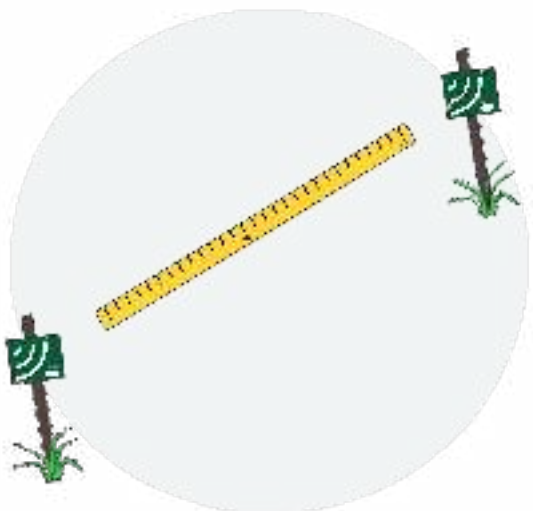
Localiser les enregistreurs



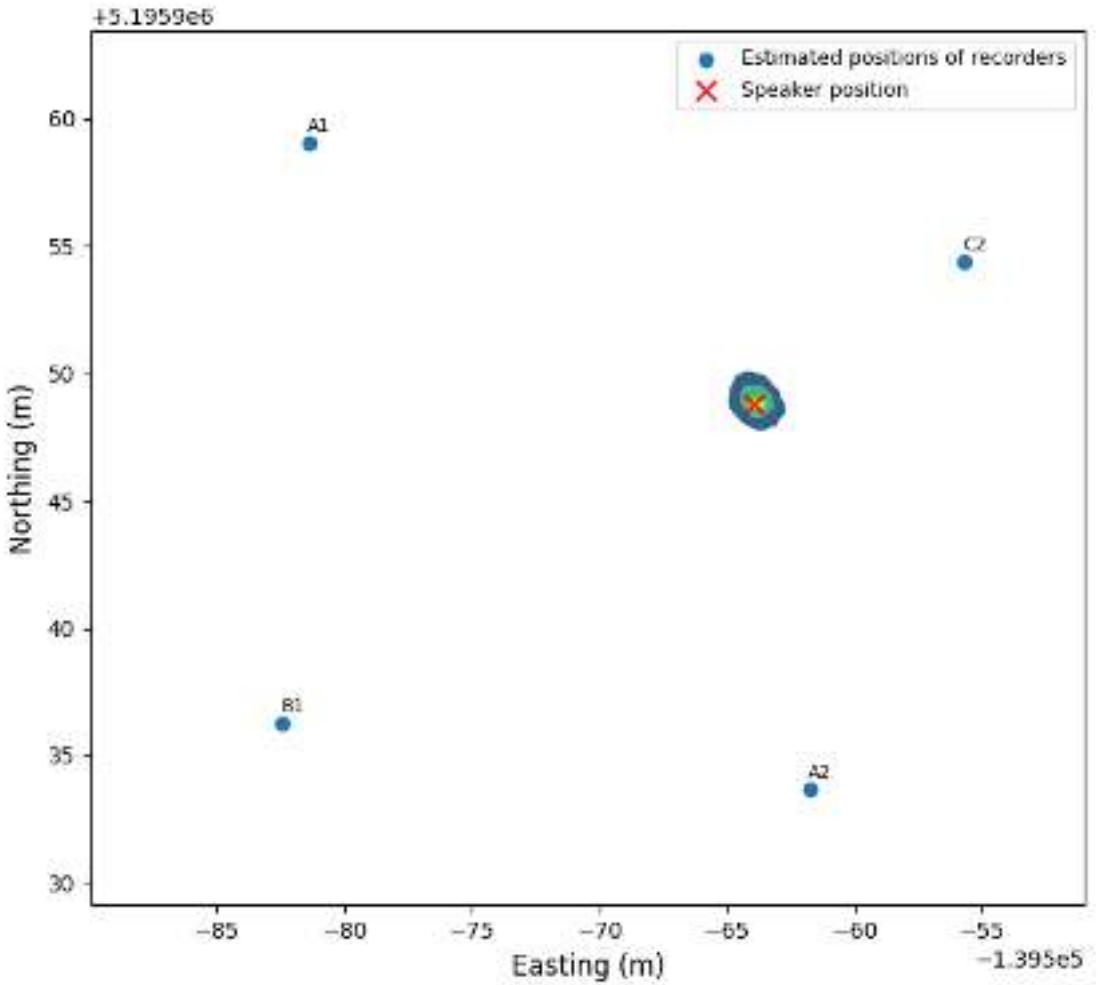
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Localiser les enregistreurs



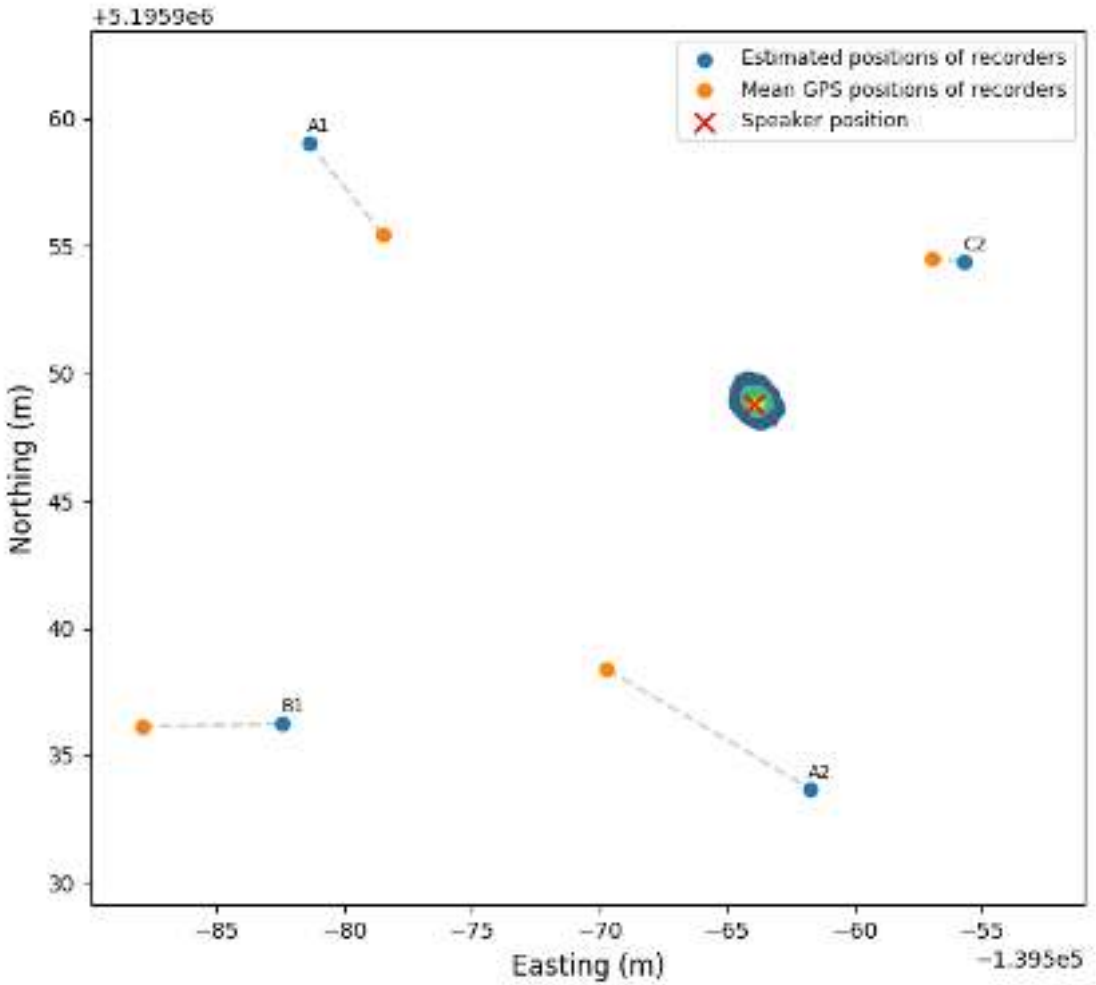


Localiser les enregistreurs



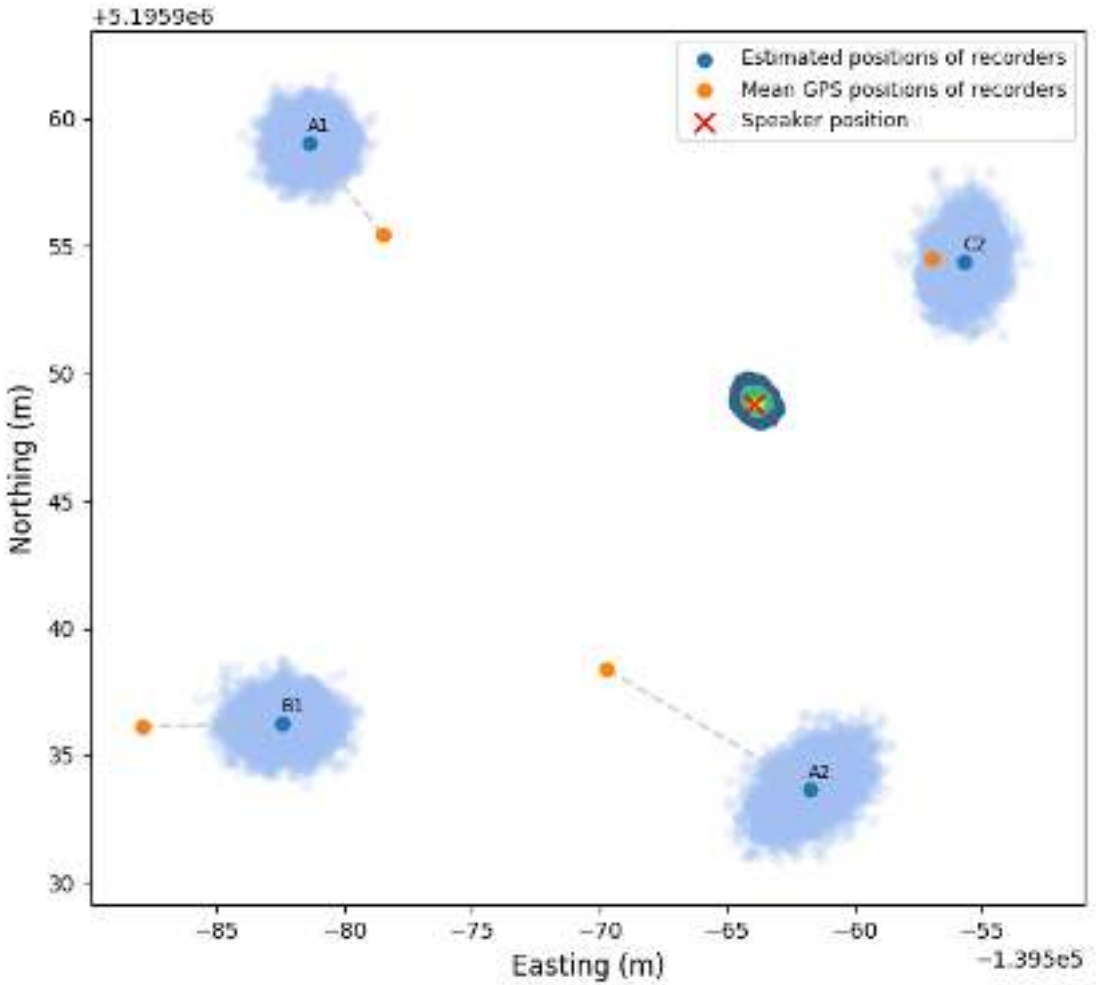
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Localiser les enregistreurs



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Localiser les enregistreurs



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Cartographier les comportements



Objectifs

✓ Localiser des sons avec précision

Prendre en compte les incertitudes:

- ✓ Positions des enregistreurs
- ✓ Estimation des délais de temps d’arrivée

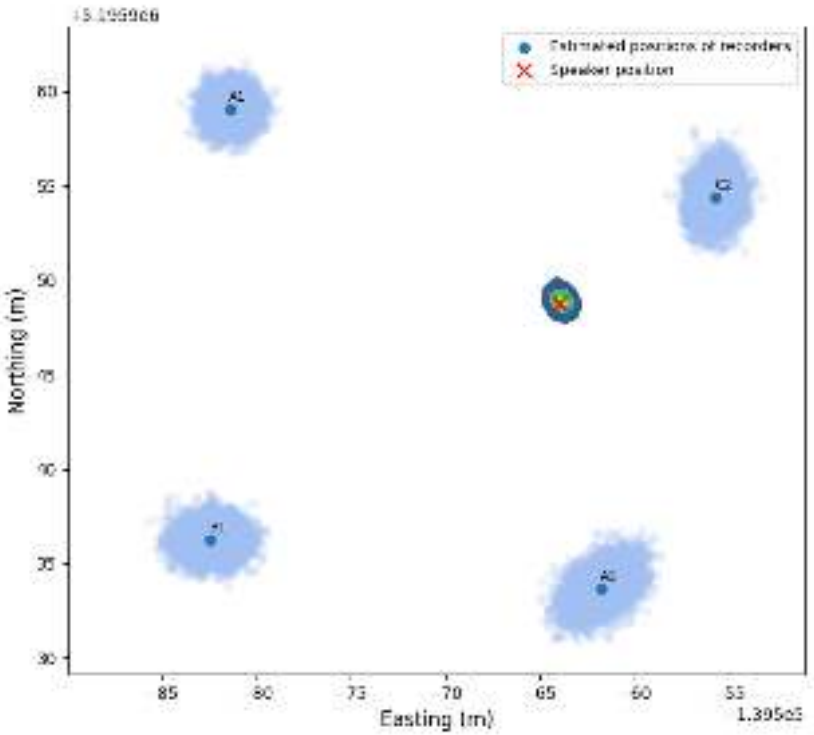


Objectifs

✓ Localiser des sons avec précision

Prendre en compte les incertitudes:

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- ✓ Estimation des délais de temps d’arrivée



Évaluer l’influence du paysage



Thank you!

Ce travail a été financé par





Contact

lucasvoirin.github.io

lucas.voirin.1@ulaval.ca

Références

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